

This document collects facts about the monic polynomial [1]

$$p_K(z) = z^{2K} - z^{2K-1} - 4z^K - z + 1, \quad K \geq 2 \text{ even}.$$

The real zero of $p_K(z)$ in $(1, 3)$ arises as the (conjectured) radius parameter maximizing the “worst case” minimum distance between nontrivially distinct Huffman sequences.¹

As we show here, $p_K(z)$ exhibits some very interesting properties. And curiously, many of these properties are paralleled by the Gonchar polynomials [2] (see the bottom of pg. 4).

Fact 1. $p_K(z)$ is self-reciprocal; if ζ is a zero of $p_K(z)$, then so is $1/\zeta$.

Proof. It holds that $z^{2K}p_K(z^{-1}) = z^{2K}(1 - z^{-1} - 4z^{-K} - z^{-2K+1} + z^{-2K}) = p_K(z)$. ■

Fact 2. $A_K(z) = -p_K(z)/z^K$ is an AACF; if ζ is a zero of $p_K(z)$, then so is $1/\zeta^*$.

Proof. We have $A_K(z) = -z^K + z^{K-1} + 4 + z^{-(K-1)} - z^{-K}$, which is clearly trigonometric. On the unit circle,

$$A_K(e^{j\omega}) = 2 \cos((K-1)\omega) - 2 \cos(K\omega) + 4,$$

and $\cos((K-1)\omega) - \cos(K\omega) \geq -2$ implies $A_K(e^{j\omega}) \geq 0$. ■

Fact 3. $p_K(z)$ has a zero of multiplicity two at $z = -1$; these are the only zeros of $p_K(z)$ lying on the unit circle.

Proof. For $\omega \in (-\pi, \pi]$, $p_K(e^{j\omega}) = 0$ is equivalent to $A_K(e^{j\omega}) = 0$, which results when

$$\cos(K\omega) - \cos((K-1)\omega) = 2 \iff \cos(K\omega) = 1 \text{ and } \cos((K-1)\omega) = -1.$$

As $K \geq 2$ is even, the only $\omega \in (-\pi, \pi]$ satisfying both conditions is $\omega = \pi$, and $e^{j\pi} = -1$. The fact that this zero has multiplicity two follows from

$$\begin{aligned} p'_K(-1) &= 2K(-1)^{2K-1} - (2K-1)(-1)^{2K-2} - 4K(-1)^{K-1} - 1 \\ &= -2K - (2K-1) + 4K - 1 \\ &= 0, \end{aligned}$$

while

$$\begin{aligned} p''_K(-1) &= 2K(2K-1)(-1)^{2K-2} - (2K-1)(2K-2)(-1)^{2K-3} - 4K(K-1)(-1)^{K-2} \\ &= 2K(2K-1) + (2K-1)(2K-2) - 4K(K-1) \\ &= (4K^2 - 2K) + (4K^2 - 6K + 2) - (4K^2 - 4K) \\ &= 4K^2 - 4K + 2 \\ &= 4K(K-1) + 2 \\ &> 0. \end{aligned}$$

This concludes our proof. ■

¹The value of this zero has an exact geometric interpretation independent of the conjecture’s validity. For even $K \geq 2$, the zero of $p_K(z)$ in $(1, 3)$ is the radius parameter for which the single-flip and alternating signed patterns yield equal phase-invariant distances (see [1]).

Fact 4. *Barring the repeated zero at $z = -1$, all zeros of $p_K(z)$ are simple.*

Proof. Suppose ζ is a repeated zero of $p_K(z)$. Then $A_K(\zeta) = A'_K(\zeta) = 0$, so let us compute

$$A'_K(z) = -Kz^{K-1} + (K-1)z^{K-2} - (K-1)z^{-K} + Kz^{-(K+1)}.$$

Equating this derivative to zero and multiplying both sides by ζ^{K+1} , we have

$$-K\zeta^{2K} + (K-1)\zeta^{2K-1} - (K-1)\zeta + K = 0 \iff \zeta^{2K-1}[K\zeta - (K-1)] = K - (K-1)\zeta.$$

For $r = |\zeta|$, the above is equivalent to

$$r^{2K-1}|K\zeta - (K-1)| = |K - (K-1)\zeta|.$$

There are three cases: $r > 1$, $r < 1$, and $r = 1$. A useful identity is

$$|K\zeta - (K-1)|^2 - |K - (K-1)\zeta|^2 = (2K-1)(r^2 - 1).$$

Assume $r > 1$. Then $|K\zeta - (K-1)| > |K - (K-1)\zeta|$ (see above identity) and $r^{2K-1} > 1$, meaning the desired equality does not hold.

Assume $r < 1$. Then $|K\zeta - (K-1)| < |K - (K-1)\zeta|$ (see above identity) and $r^{2K-1} < 1$, so the desired equality fails to hold.

Hence, it must be that $r = 1$. But we know from Fact 3 that $\zeta = -1$ is the only zero of $p_K(z)$ with $|\zeta| = 1$. All other zeros of $p_K(z)$ are therefore simple. ■

Fact 5. *$p_K(z)$ has four real zeros, namely, $R_K \in (1, 3)$, $1/R_K \in (1/3, 1)$, and two zeros (counting multiplicity) at $z = -1$.*

Proof. Suppose $p_K(z)$ has a real zero $R_K = e^t \geq 1$, $t \geq 0$. This requires

$$A_K(e^t) = 0 \iff \cosh(Kt) - \cosh((K-1)t) = 2 \iff \sinh\left(\frac{(2K-1)t}{2}\right) \sinh\left(\frac{t}{2}\right) = 1.$$

Consider the function

$$f : [0, \infty) \rightarrow \mathbb{R}, \quad f(t) = \sinh\left(\frac{(2K-1)t}{2}\right) \sinh\left(\frac{t}{2}\right) - 1.$$

Since $f(0) = -1$, $e^0 = 1$ is not a zero of $p_K(z)$. But $f(t)$ is strictly increasing for $t > 0$, so there exists a $t_K > 0$ with $f(t_K) = 0$. Hence $p_K(z)$ has a unique zero $R_K = e^{t_K} > 1$. Indeed, for all $K \geq 2$, $p_K(1) = -1$ and $p_K(3) \geq 16$. Thus the IVT implies $R_K \in (1, 3)$. By self-reciprocity (see Fact 1), $1/R_K \in (1/3, 1)$ is the only zero of $p_K(z)$ in $(0, 1)$.

Now suppose $p_K(z)$ has a real zero $r_K = -e^t < -1$, $t > 0$. This requires

$$A_K(-e^t) = 0 \iff \cosh(Kt) + \cosh((K-1)t) = 2.$$

But $\cosh(t) > 1$ for all $t > 0$, so the above equation has no solution. We conclude that $p_K(z)$ has no zeros in the interval $(-\infty, -1)$, and by self-reciprocity, no zeros in $(-1, 0)$.

We already showed $p_K(z)$ has a double zero at $z = -1$ (see Fact 3), and since $p_K(0) \neq 0$, we are done. ■

Fact 6. $R_2 = \varphi^2$ and $R_4 = \rho^2$, where φ and ρ are the golden and plastic ratios, respectively.

Proof. We have $p_2(z) = z^4 - z^3 - 4z^2 - z + 1 = (z+1)^2(z^2 - 3z + 1)$, and $\varphi^2 = \varphi + 1$. Hence

$$\begin{aligned} p_2(\varphi^2) &= (\varphi^2 + 1)^2[(\varphi + 1)^2 - 3(\varphi + 1) + 1] \\ &= (\varphi^2 + 1)^2[(\varphi + 1) + 2\varphi + 1 - 3(\varphi + 1) + 1] \\ &= 0. \end{aligned}$$

Also, $p_4(z) = z^8 - z^7 - 4z^4 - z + 1 = q(z)(z^3 - 2z^2 + z - 1)$, $q(z) = (z+1)^2(z^3 - z^2 + 2z - 1)$. Since $\rho^3 = \rho + 1$, we get

$$\begin{aligned} p_4(\rho^2) &= q(\rho^2)[(\rho + 1)^2 - 2(\rho + 1)\rho + \rho^2 - 1] \\ &= q(\rho^2)[(\rho^2 + 2\rho + 1) - 2(\rho^2 + \rho) + \rho^2 - 1] \\ &= 0. \end{aligned}$$

Done. ■

Fact 7. The sequence of zeros $\{R_K\}$, $K \geq 2$ even, is strictly decreasing with limit $L = 1$.

Proof. We adopt an approach similar to the proof of [3, Prop. 1]. To begin, observe that $p_K(z) = 0$ requires

$$A_K(z) = 0 \iff 4 + z^{K-1} + z^{-(K-1)} = z^K + z^{-K} \iff \frac{4 + z^{K-1} + z^{-(K-1)}}{z^K + z^{-K}} = 1.$$

For $z \in (1, 3)$, consider the function

$$h : [1, \infty) \rightarrow \mathbb{R}, \quad h(K) = \frac{4 + z^{K-1} + z^{-(K-1)}}{z^K + z^{-K}}.$$

The derivative of $h(K)$ is

$$h'(K) = -\frac{2z^{K-1} \log(z)(2z^{2K-1} + z^K(z^2 - 1) - 2z)}{(z^{2K} + 1)^2}.$$

Since $h'(K) < 0$ for all $z \in (1, 3)$, $h(K)$ is strictly decreasing in K . So for any $R_K > 1$, $K \geq 2$ even,

$$\frac{4 + R_K^{K-1} + R_K^{-(K-1)}}{R_K^K + R_K^{-K}} = 1 \quad \text{and} \quad \frac{4 + R_K^{K+1} + R_K^{-(K+1)}}{R_K^{K+2} + R_K^{-(K+2)}} < 1,$$

where the right inequality is equivalent to $p_{K+2}(R^K) > 0$. Indeed, R_{K+2} is the only zero of $p_{K+2}(z)$ in $(1, 3)$, and $p_{K+2}(z) > 0$ for $z \geq 3$ (see Fact 5). It follows that $R_{K+2} < R_K$.

Now note that because $\{R_K\}$ is bounded and strictly decreasing, it converges to a limit $L \geq 1$. Let us assume that $L > 1$. Then for any $R_K > L$, we have

$$p_K(R^K) = 0 \iff (R_K - 1)(R_K^{2K-1} - 1) = 4R_K^K \iff (R_K - 1)(R_K^{K-1} - R_K^{-K}) = 4.$$

In particular, $(L - 1)(L^{K-1} - L^{-K}) \leq (R_K - 1)(R_K^{K-1} - R_K^{-K})$, and

$$\lim_{K \rightarrow \infty} (L - 1)(L^{K-1} - L^{-K}) = \infty.$$

But this is a contradiction, so we conclude that $L = 1$. ■

Fact 8. *The zeros of $p_K(z)$ lie in the closed annulus $\{z \in \mathbb{C} : 1/R_K \leq |z| \leq R_K\}$.*

Proof. If $p_K(\zeta) = (\zeta - 1)(\zeta^{2K-1} - 1) - 4\zeta^K = 0$, then

$$|(\zeta - 1)(\zeta^{2K-1} - 1)| = 4|\zeta|^K.$$

Suppose $r = |\zeta| \geq 1$. By the reverse triangle inequality, there holds

$$(r - 1)(r^{2K-1} - 1) \leq 4r^K.$$

Thus $(r^K + r^{-K}) - (r^{K-1} + r^{-(K-1)}) \leq 4$, and substituting $r = e^t$ with $t \geq 0$ yields

$$\sinh\left(\frac{(2K-1)t}{2}\right) \sinh\left(\frac{t}{2}\right) \leq 1.$$

We have equality iff $t = t_K$ (see Fact 5), implying that if $p_K(\zeta) = 0$, then $|\zeta| \leq R_K = e^{t_K}$. Now suppose $r = |\zeta| \leq 1$. By self-reciprocity (see Fact 1), $1/\zeta$ is a zero of $p_K(z)$ with $1/|\zeta| \geq 1$. In particular, $1/|\zeta| \leq R_K$ so that $|\zeta| \geq 1/R_K$. ■

Fact 9. *The zeros of $p_K(z)$ are uniformly distributed in phase as $K \rightarrow \infty$.*

Proof. For $0 \leq \alpha < \beta \leq 2\pi$, let $N(\alpha, \beta)$ denote the number of zeros of $p_K(z)$ with phase (modulo 2π) in the interval $[\alpha, \beta]$. Invoking [4, Thm. 1], we have

$$\left| \frac{N(\alpha, \beta)}{2K} - \frac{\beta - \alpha}{2\pi} \right| < 16\sqrt{\frac{\log(8)}{2K}}.$$

Since $\log(8)/2K \rightarrow 0$ as $K \rightarrow \infty$, it must be that $N(\alpha, \beta)/2K \rightarrow (\beta - \alpha)/2\pi$. ■

Fact 10. *The nonreal zeros of $p_K(z)$ occur in quadruples $\{\zeta, \zeta^*, 1/\zeta, 1/\zeta^*\}$; there are exactly $(K - 2)/2$ such quadruples.*

Proof. Suppose ζ is a nonreal zero of $p_K(z)$. Since $p_K(z)$ has real coefficients, ζ^* is also a zero of $p_K(z)$. Furthermore, Fact 1 and Fact 2 imply, respectively, that $p_K(z)$ has the zeros $1/\zeta$ and $1/\zeta^*$. These four zeros are distinct by Fact 3 and the assumption that ζ is nonreal. Counting multiplicities, $p_K(z)$ has $2K$ zeros. Four of these are real (see Fact 5), so there are

$$2K - 4 = 4\left(\frac{K - 2}{2}\right)$$

nonreal zeros, which occur in exactly $(K - 2)/2$ quadruples. ■

As noted in [1], the polynomials $\{p_K(z) : K \geq 2 \text{ even}\}$ have very similar properties to the Gonchar polynomials [2]. The K th Gonchar polynomial is given by

$$g_K(z) = [(z - 1)^K - z - 1]z^{K-1} + (z - 1)^K, \quad K \geq 1.$$

For even K , $g_K(z)$ has three real zeros: one at $z = -1$, one in $(1/3, 1/2)$, and one in $(2, 3)$ (cf. Fact 5). The value of the zero in $(2, 3)$ is $1 + \varphi$ and $1 + \rho$ for $K = 2$ and $K = 4$, respectively (cf. Fact 6). Further, the zero in $(2, 3)$ is strictly decreasing in K (cf. Fact 7). Though $p_2(z) = (z + 1)g_2(z)$, the general connection between $p_K(z)$ and $g_K(z)$ is unclear. For reference, we plot the zeros of $p_K(z)$ and $g_K(z)$ in Fig. 1 for $K \in \{2, 4, 6, \dots, 64\}$.

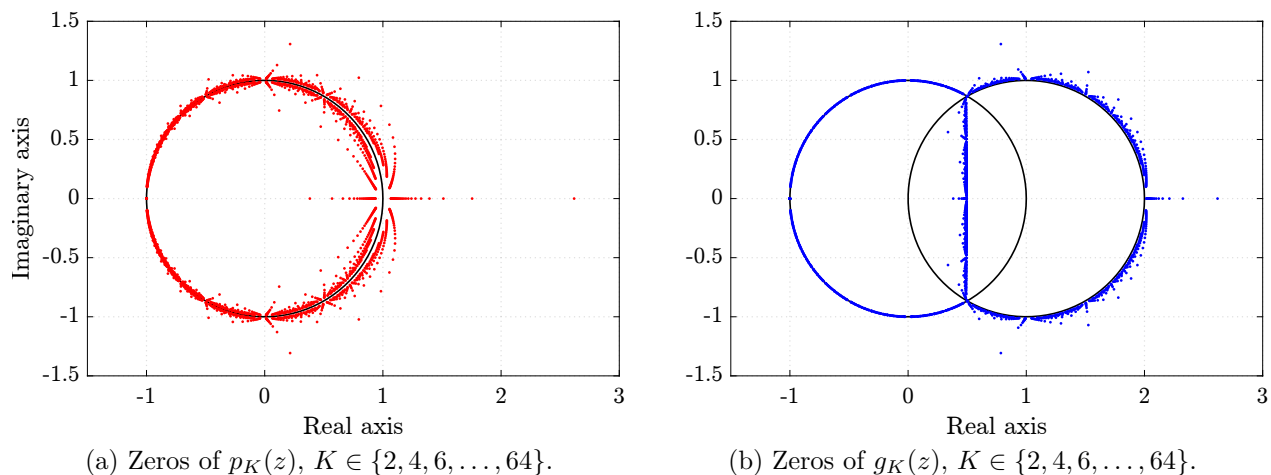


Figure 1: Complex zeros of $p_K(z)$ and $g_K(z)$ for various even values of $K \geq 2$.

References

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